

BESTYAHSKAYA ZVEROFERMA, Russian Federation

WMO: 244490

Lat: **65.300N**Lon: **124.166E**Elev: **73**StdP: **100.45**Time Zone: **9.00 (E09)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-51.0	-49.3	-52.5	0.0	-50.6	-50.6	0.0	-49.5	4.5	-24.3	4.2	-25.0	0.0	310	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	27.6	18.0	25.7	17.2	23.8	16.2	19.6	25.3	18.4	23.9	17.2	22.2	2.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.6	22.2	16.3	11.7	21.0	14.8	10.6	20.2	56.6	25.9	52.7	23.7	48.7	22.3	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.1	4.3	3.8	DB	-51.6	30.9	2.2	1.3	-53.2	31.8	-54.5	32.5	-55.8	33.3	-57.4	34.2	(4)
(5)			WB	-51.6	21.5	2.2	1.1	-53.2	22.3	-54.5	22.9	-55.8	23.5	-57.4	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.9	-38.1	-34.0	-19.4	-6.1	3.9	14.3	15.4	13.3	4.3	-8.5	-28.6	-37.0	(6)	
	DBStd	20.71	7.16	6.89	8.49	6.30	5.00	4.45	3.73	3.71	4.17	6.33	6.90	6.70	(7)	
	HDD10.0	7707	1491	1231	912	483	197	12	3	13	177	575	1157	1457	(8)	
	HDD18.3	10340	1749	1464	1170	733	447	134	106	159	421	833	1407	1716	(9)	
	CDD10.0	439	0	0	0	0	8	140	171	114	6	0	0	0	(10)	
	CDD18.3	31	0	0	0	0	1	12	16	2	0	0	0	0	(11)	
	CDH23.3	472	0	0	0	0	5	218	199	49	0	0	0	0	(12)	
	CDH26.7	89	0	0	0	0	1	46	39	2	0	0	0	0	(13)	
(14) Wind	WSAvg	1.6	1.1	1.3	1.5	1.9	2.0	1.8	1.9	1.7	1.6	1.7	1.3	1.3	(14)	
(15) Precipitation	PrecAvg	387	16	14	17	19	32	46	51	56	43	49	26	18	(15)	
	PrecMax	549	34	35	52	40	58	91	99	132	81	98	57	46	(16)	
	PrecMin	296	6	2	3	5	9	8	12	11	10	15	11	9	(17)	
	PrecStd	58	6	8	10	10	11	18	20	26	19	19	12	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-17.7	-14.2	2.7	9.3	22.9	30.0	29.9	26.8	20.7	3.6	-7.7	-20.0	(19)	
		MCWB	-18.0	-14.6	-0.3	4.5	11.3	19.2	19.7	17.8	14.1	2.0	-8.6	-20.2	(20)	
	2%	DB	-21.4	-17.3	-1.0	7.0	19.1	27.9	27.7	24.8	17.0	1.8	-11.3	-21.9	(21)	
		MCWB	-21.6	-17.7	-3.0	2.7	10.6	17.3	19.1	17.0	11.1	0.7	-11.7	-22.1	(22)	
	5%	DB	-24.5	-21.5	-4.0	5.4	16.3	26.0	25.7	22.7	14.1	0.5	-14.3	-23.5	(23)	
		MCWB	-24.7	-21.8	-5.4	1.4	9.1	16.6	18.1	16.5	9.7	-0.4	-14.6	-23.7	(24)	
	10%	DB	-27.6	-23.8	-6.1	3.7	13.7	23.9	23.7	20.8	11.7	-0.8	-18.3	-25.9	(25)	
		MCWB	-27.7	-24.1	-7.2	0.5	7.8	15.5	16.9	15.5	8.0	-1.7	-18.7	-26.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-17.9	-14.9	-0.1	4.7	12.7	20.1	21.2	20.1	14.7	2.6	-8.7	-20.1	(27)	
		MCDB	-17.8	-14.4	2.1	8.7	20.9	28.3	26.5	25.0	20.9	3.1	-8.1	-19.9	(28)	
	2%	WB	-21.7	-17.6	-3.0	3.1	10.7	18.8	20.1	18.4	11.7	1.0	-11.6	-22.2	(29)	
		MCDB	-21.5	-17.2	-1.3	6.2	17.6	25.9	25.6	22.5	15.5	1.6	-11.3	-22.0	(30)	
	5%	WB	-24.6	-21.8	-5.3	1.8	9.7	17.5	19.1	17.1	10.2	-0.2	-14.7	-23.7	(31)	
		MCDB	-24.5	-21.6	-4.0	4.6	16.4	24.0	24.2	21.1	13.7	0.6	-14.3	-23.4	(32)	
	10%	WB	-27.7	-24.0	-7.1	0.8	8.0	16.2	17.8	16.1	8.7	-1.7	-18.5	-26.1	(33)	
		MCDB	-27.5	-23.8	-5.9	3.3	13.3	23.0	23.1	20.4	11.4	-0.9	-18.2	-25.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	10.2	13.1	12.7	10.4	12.3	11.4	11.4	10.2	6.9	8.4	6.7	(35)	
		MCDBR	8.4	10.3	13.1	12.4	16.4	16.9	15.2	15.8	14.9	6.3	8.3	8.2	(36)	
		MCWBR	8.3	10.0	11.5	9.0	9.2	8.2	7.7	8.9	9.6	5.1	8.2	8.1	(37)	
	5% WB	MCDBR	8.4	10.3	12.4	10.6	14.4	14.7	12.2	13.4	13.3	5.5	8.5	8.1	(38)	
		MCWBR	8.3	9.9	11.0	7.8	8.8	7.6	6.4	8.2	9.0	4.7	8.4	8.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.186	0.243	0.288	0.334	0.372	0.391	0.480	0.472	0.325	0.261	0.185	0.182	(40)	
	taud		1.875	2.064	2.085	2.087	2.260	2.343	2.047	2.013	2.450	2.245	1.897	1.881	(41)	
	Ebn at Noon		296	639	772	821	828	820	714	668	752	619	282	67	(42)	
	Edh at Noon		23	59	94	122	114	108	142	133	68	49	20	7	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.14	0.95	2.77	4.93	5.49	5.88	5.37	3.82	2.29	0.82	0.26	0.04	(44)	
	RadStd		0.01	0.04	0.12	0.21	0.39	0.43	0.34	0.41	0.26	0.12	0.02	0.00	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page